Learning ^{1}H and ^{13}C NMR chemical shifts jointly from unassigned 2-D HSQC peak sets

A sliced sort-match weak-supervision recipe, causally audited and calibrated for dereplication

Zhaohe Dong[1,*]

[1]Yusuf Hamied Department of Chemistry, University of Cambridge, Lensfield Road, Cambridge CB2 1EW, United Kingdom.

[*]Correspondence: zd314@cam.ac.uk

April 2026

Abstract

Background. Nuclear magnetic resonance (NMR) chemical-shift predictors are almost always trained on atom-assigned spectra, where every training example specifies which atom in the molecular graph produced which peak. Atom assignment is the expensive part of curating NMR data, so most published spectra never enter a training set. Two-dimensional HSQC peak lists are a different matter: they appear in nearly every modern organic paper, pair the ^{1}H and ^{13}C shifts of directly bonded H–C groups, and carry no atom identity. Concurrent work by Jin et al. [11] has established that such unassigned HSQC peak sets can be used as permutation-invariant set supervision for joint ^{1}H / ^{13}C prediction, via exact Hungarian bipartite matching at million-spectrum scale. We address four remaining questions that are specifically relevant to small-scale, chemistry-driven deployment and that Jin et al. do not report on: (i) whether the predictor can be wrapped with a rigorous molecule-level uncertainty guarantee suitable for natural-product dereplication; (ii) whether the weakly supervised ^{1}H head is actually learning from the HSQC signal rather than from encoder leakage out of the supervised ^{13}C loss; (iii) whether a cheaper sliced-Wasserstein loss can replace the $O(n^3)$ exact Hungarian step at small scale; and (iv) whether the predictor can discriminate correct structures from constitutional isomers well enough to function as a dereplication ranker.

Results. We train a four-layer graph isomorphism network on a 1,542-molecule NMRShiftDB2 subset with 10% atom-assigned ^{13}C labels and HSQC peak lists for the remaining 90%, treating each HSQC peak list as an unordered multiset in (δ_H, δ_C) space. The headline test MAE is 4.53 ± 0.11 ppm ^{13}C and 0.35 ± 0.02 ppm ^{1}H over three seeds, with no atom-assigned ^{1}H labels ever used; layered on full ^{13}C supervision the same recipe reaches 3.23 ± 0.10 ppm ^{13}C and 0.30 ± 0.03 ppm ^{1}H. On top of this headline, four experimental deliverables establish the paper's specific contributions in canonical order: (i) **Bonferroni-corrected split-conformal calibration** delivers a rigorous 94.8% molecule-level coverage guarantee at $\alpha = 0.05$, alongside per-atom coverage of 95.2% / 96.7%; (ii) a **causal ^{1}H-zeroing audit**, in which the ^{1}H coordinate of every HSQC target is zeroed and the model is retrained, collapses ^{1}H error to 4.69 ± 0.10 ppm

while leaving ^{13}C essentially at its baseline, confirming the ^{1}H head learns from the HSQC signal rather than from gradient leakage out of the ^{13}C head; (iii) the underlying **sliced sort-match loss** reduces the 2-D optimal-transport problem to K differentiable 1-D sort problems at $O(K\,n \log n)$ per molecule, asymptotically cheaper than the $O(n^3)$ exact Hungarian update used by Jin et al. [11] and batchable through `torch.sort` without a custom autograd hack; and (iv) against constitutional isomers the model is correct on 77% of candidates and wrong on 22%, a 3.5-fold dereplication-ranking ratio explicitly flagged as insufficient for standalone accept/reject.

Conclusions. Complementary to the feasibility established concurrently by Jin et al. [11], we contribute four technical advances targeted at small-scale, chemistry-driven deployment, in order: (i) a Bonferroni-corrected molecule-level split-conformal coverage guarantee for NMR shift prediction, of which we are not aware of a prior example in the literature; (ii) a causal audit of the ^{1}H supervision pathway that rules out gradient leakage from the ^{13}C head; (iii) an $O(K\,n \log n)$ sliced-Wasserstein loss that replaces the $O(n^3)$ exact Hungarian update used by [11]; and (iv) an explicit constitutional-isomer discrimination experiment for natural-product dereplication ranking. The validation in this paper uses HSQC multisets **derived from NMRShiftDB2 atom-assigned spectra**, not scraped literature peak lists; carrying the pipeline to real literature HSQC data is the single largest outstanding gap. Code, data filters, trained checkpoints, and an artefact map are at `https://github.com/dongzhaohe321418-lab/nmr-ssl` (release `v2.0-jin-revision`).

Keywords: NMR chemical shift prediction; semi-supervised learning; sliced Wasserstein; optimal transport; split-conformal prediction; natural-product dereplication; graph neural network.

Background

Predicting NMR chemical shifts from a molecular graph is a standard machine-learning task in chemoinformatics. The current best predictor, NMRNet [1], reaches 1.10 ppm ^{13}C and 0.18 ppm ^{1}H mean absolute error on NMRShiftDB2. It achieves this by training two separate heads on two separately-curated atom-assigned corpora. Every training example carries per-atom labels that someone had to annotate.

Per-atom annotation is the slow, expensive part of building an NMR training set. Most spectra in the literature never get annotated to that level, so they cannot be used with the standard supervised recipe. 2-D experiments such as HSQC, HMBC, COSY and NOESY appear in far more published papers than fully atom-assigned 1-D spectra, but their peak lists are multisets of correlations rather than atom-indexed values, which rules them out of conventional training.

HSQC is the smallest step away from 1-D supervision. Each cross-peak in an HSQC peak list pairs a ^{1}H shift with the ^{13}C shift of its directly-bonded carbon. The pairing is known; the atom identity is not. Jin et al. [11], in concurrent work posted on arXiv in January 2026, have established that this form of weakly-paired supervision is in fact rich enough to train a joint ^{1}H / ^{13}C predictor at scale, using an exact Hungarian bipartite matching loss over millions of literature-extracted spectra. Their feasibility result is not in dispute. The present paper addresses four remaining questions that are specifically relevant to small-scale, chemistry-driven deployment, and that Jin et al. do not report on.

Related work and contributions of this paper

We do not claim priority on the unassigned-HSQC-as-supervision formulation; that priority belongs to Jin et al. [11]. What we contribute is a set of four complementary technical and statistical advances, each of which targets a failure mode of the exact-Hungarian, large-scale recipe.

First — a formal molecule-level coverage guarantee. We wrap the predictor with split-conformal calibration [4, 5] and, using a Bonferroni correction over the per-molecule HSQC peak count, recover a rigorous molecule-level coverage guarantee. Empirical per-atom test coverage at $\alpha = 0.05$ is 95.2% on ^{13}C and 96.7% on ^{1}H; the Bonferroni-corrected joint pass rate lands at **94.8**%, within 0.2% of the nominal target. To our knowledge this is the first NMR chemical-shift predictor that ships with a formal molecule-level uncertainty guarantee. Jin et al. [11] report point predictions only.

Second — a causal audit of the supervision pathway. A shared graph encoder with two output heads is vulnerable to a well-known confound: the ^{1}H head could be learning from gradient leakage out of the ^{13}C supervised loss rather than from the HSQC target itself. We rule this out by zeroing the ^{1}H coordinate of every HSQC training target and retraining with the same hyperparameters. The ^{1}H test MAE jumps from 0.35 ± 0.02 ppm to 4.69 ± 0.10 ppm, more than ten times worse than the baseline, while the ^{13}C error barely moves. The HSQC ^{1}H coordinate is doing the work. Jin et al. [11] do not report this ablation.

Third — a sliced-Wasserstein alternative to exact Hungarian matching. Across both coordinate axes jointly we use the sliced-Wasserstein construction of Bonneel et al. [3], which averages a 1-D sort-match loss over K random linear projections and brings the per-molecule cost from $O(n^3)$ (exact Hungarian, as in [11]) down to $O(K\, n \log n)$. Along either coordinate

axis the inner 1-D loss is exactly minimized by sort-matching, by the Hardy– Littlewood–Pólya rearrangement inequality [6]; we give a full statement and proof of this 1-D sort-match theorem in Supplementary Information **S8**, so that the core methodological result of this paper is self-contained. The sliced construction is permutation-invariant, differentiable through `torch.sort`, and batchable without a custom autograd hack — useful properties in low-compute regimes where the Hungarian inner loop of [11] would dominate training time.

Fourth — an explicit dereplication experiment. We evaluate the predictor against three levels of wrong-candidate controls — random pair, scaffold neighbour, and constitutional isomer — with the hardest, chemistry-meaningful setting being the constitutional isomer control (same molecular formula, different connectivity). The correct-candidate joint pass rate is 77.0% and the wrong-candidate joint pass rate is 21.7%, giving a 3.5-fold discrimination ratio. This is explicitly positioned as a candidate ranker for natural-product dereplication pipelines, not as a standalone binary accept/reject classifier. Jin et al. [11] do not report an isomer-discrimination experiment.

Boundary of the claim. All experiments in this paper use HSQC multisets **derived from NMRShiftDB2 atom-assigned spectra**, not from scraped literature peak lists; we construct the multiset by averaging the attached-^{1}H shifts at each H-bearing carbon. This preserves the combinatorial structure of a real HSQC peak list but not its experimental noise characteristics, and we flag the literature-scraping step as the single largest outstanding gap. Headline label-efficiency numbers in this paper are reached at the 1,542-molecule NMRShiftDB2 filtered-subset scale, not at the literature scale reported by Jin et al. [11]; absolute accuracy is correspondingly behind and we do not claim otherwise.

The rest of the paper is organized as follows. *Methods* writes out the sliced sort-match loss and its theoretical motivation, pointing to Supplementary **S8** for the full statement and proof of the underlying 1-D sort-match theorem. *Experimental details* covers dataset construction, model architecture, training protocol, split-conformal calibration, and the structure-verification procedure. *Results* reports the main label-efficiency numbers, the causal ^{1}H-zeroing audit, the combined supervision regime, the label-efficiency sweep, the three wrong-candidate controls, the scaffold-OOD split, robustness to realistic HSQC degradation, the hyperparameter and axis-aligned-decomposition sweeps, the per-carbon error decomposition, and the split-conformal coverage analysis. *Discussion* covers honest limitations and the path to real literature HSQC data. *Conclusions* summarizes the findings with explicit reference to Jin et al. [11] as the concurrent state-of-the-art feasibility result. The Supplementary Information contains per-seed tables, hyperparameter sweep data, a stop-gradient control, a pretrain-then-finetune baseline, the stereo-rich stress test, a multiplicity-augmented loss experiment, and the full sort-match theorem (S8).

Methods

Set-valued NMR supervision

Fix a molecule M with its molecular graph $G(M)$ and write $\mathbf{P}^\star(M) = \{(\delta^\star_{H,j}, \delta^\star_{C,j})\}_{j=1}^n$ for its observed HSQC peak set. The target is an unordered multiset: the training pipeline sees the n pairs but not which pair corresponds to which heavy atom. A graph neural network $f_\theta : G \to (\hat{\delta}_H, \hat{\delta}_C)$ predicts per-atom shifts; from its outputs at the H-bearing carbons we form

the predicted set $\hat{\mathbf{P}}(M,\theta) = \{(\hat{\delta}_{H,i}, \hat{\delta}_{C,i})\}_{i=1}^{n}$.

The goal is to minimize a permutation-invariant loss between $\hat{\mathbf{P}}$ and $\mathbf{P}^\star$. In one dimension this reduces exactly to sorting both sets and taking a per-element ℓ^2 distance, as a direct consequence of the Hardy–Littlewood–Pólya rearrangement inequality [6]; we give a full statement and proof of this 1-D sort-match theorem in Supplementary **S8**, so that the methodological backbone of this paper is self-contained within the manuscript. In two dimensions the exact solution is Hungarian optimal bipartite matching — the approach taken by Jin et al. [11] — which is $O(n^3)$ per molecule and not easily batched on GPU.

Sliced approximation

We follow Bonneel et al. [3] and use the sliced-Wasserstein distance as a cheap upper bound. For a random direction $\theta \in S^1$ drawn uniformly from the unit sphere, define the projection $\Pi_\theta(h,c) = h\cos\theta + c\sin\theta$. Then

$$SW_2^2(\hat{\mathbf{P}}, \mathbf{P}^\star) = \mathbb{E}_{\theta\sim\mathrm{Unif}(S^1)}\big[W_2^2(\Pi_\theta\hat{\mathbf{P}}, \Pi_\theta\mathbf{P}^\star)\big],$$

where each inner one-dimensional 2-Wasserstein is solved exactly by sorting the two projected sets. A Monte-Carlo estimator with K directions gives the **sliced sort-match loss**,

$$\mathcal{L}_{\mathrm{SSW}}(\hat{\mathbf{P}}, \mathbf{P}^\star) = \frac{1}{K}\sum_{k=1}^{K}\mathcal{L}_{\mathrm{sort}}(\Pi_{\theta_k}\hat{\mathbf{P}}, \Pi_{\theta_k}\mathbf{P}^\star),$$

which is permutation-invariant, differentiable almost everywhere through `torch.sort`, and computed in $O(Kn\log n)$ per molecule. It is a consistent estimator of SW_2^2 as $K\to\infty$, and satisfies $SW_2^2 \le W_2^2$ by the classical inequality.

Numerical audit. On 20 random two-dimensional point-set pairs drawn from the empirical shift distribution of NMRShiftDB2 ($n \in [5,20]$, $K = 64$), the ratio $\mathcal{L}_{\mathrm{SSW}}/W_2^{\mathrm{Hungarian}}$ concentrates in $[0.30, 0.54]$ with mean 0.41 ± 0.06, consistent with the expected two-dimensional bound. Implementation: `src/nmr2d/losses_2d.py`. Verification test: `tests/test_theorem_2d.py`.

Axis-aligned decomposition

A specialization of the sliced estimator uses only the two native coordinate axes (i.e. $\theta \in \{0, \pi/2\}$). This reduces to two independent one-dimensional sort-match losses on the ^{1}H and ^{13}C projections, with no Monte-Carlo randomness. At equal K it is a biased lower bound on the full sliced estimator, but on NMRShiftDB2 it matches sliced $K = 16$ within noise at one-eighth the cost (see *Results: Hyperparameter sensitivity*). We report both variants; the sliced-random construction is the cleaner theoretical scaffolding and preserves the inequality $SW_2^2 \le W_2^2$, while the axis-aligned variant is the cheaper form and is what we use in the structure-verification experiments.

Full training objective

The full training loss adds a supervised ^{13}C mean squared error on the labelled split:

$$\mathcal{L}(\theta) = \underbrace{\frac{1}{|\mathcal{L}|}\sum_{M\in\mathcal{L}}\frac{1}{|C(M)|}\sum_{a\in C(M)}\left(\hat{\delta}_C(a;\theta)-\delta^{\star}_C(a)\right)^2}_{\text{supervised }^{13}\text{C MSE on labelled molecules}} + \lambda\underbrace{\frac{1}{|\mathcal{U}|}\sum_{M\in\mathcal{U}}\mathcal{L}_{\mathrm{SSW}}(\hat{\mathbf{P}}(M;\theta),\mathbf{P}^{\star}(M))}_{\text{sliced sort-match on unlabelled molecules}},$$

where $\mathcal{L}$ and $\mathcal{U}$ are the labelled and unlabelled training partitions and $C(M)$ is the set of carbon atoms in molecule M. The SSL weight λ, the number of sliced directions K, and the labelled fraction are the three hyperparameters we sweep in *Results*.

Experimental details

Dataset construction

We parse the NMRShiftDB2 [9] dump `nmrshiftdb2withsignals.sd` (SourceForge, release 2026-03-15, CC-BY-SA) with RDKit [8] and join the ^{13}C and ^{1}H spectra by molecule ID. Retention filters: (a) non-degenerate ^{13}C assignments (one peak per carbon atom), (b) at least one ^{1}H assignment grouped by heavy atom, (c) at least three HSQC cross-peaks, (d) at most 60 atoms. Out of 20,000 ^{13}C records and 18,169 ^{1}H records, 1,542 molecules pass all filters.

For each retained molecule we construct the HSQC peak multiset by iterating over all H-bearing carbons and emitting the tuple $(\bar{\delta}_H, \delta_C)$, where $\bar{\delta}_H$ is the mean shift over all H atoms bonded to that carbon. This is equivalent to the multiset a single multiplicity-averaged HSQC experiment reports. Code: `src/nmr2d/data_2d.py`.

Model

A shared 4-layer graph isomorphism network (GIN [7]) encoder with dense adjacency and 20 atom features (element one-hot, degree, formal charge, aromaticity, hybridization, hydrogen count, ring membership, atomic mass), followed by two 2-layer MLP readout heads: one for per-atom ^{13}C shift and one for per-heavy-atom mean ^{1}H shift. Hidden size 192. Total parameters are approximately 340k. Code: `src/nmr2d/model_2d.py`.

Sliced sort-match loss

At each forward pass we draw K direction vectors uniformly on S^1 from a 2-D normal rescaled to unit norm, project every predicted and target peak onto each direction, sort the two projected sets within each mini-batch row, and compute the masked mean squared error on the aligned pairs. The final loss is the mean over K directions. Permutation invariance is inherited from the sort. Differentiability through `torch.sort` is standard. Code: `src/nmr2d/losses_2d.py`. The axis-aligned variant fixes $K = 2$ directions $(1, 0)$ and $(0, 1)$ and returns the sum of the two 1-D sort-match losses.

Training protocol

AdamW, learning rate 10^{-3}, weight decay 10^{-5}, batch size 32, 30 epochs, gradient clipping L2 = 5, best-val-^{13}C-MAE early stopping. Three seeds (0, 1, 2) with independent 80/10/10 random train/val/test splits and independent labelled/unlabelled partitions (labelled fraction 0.1 unless otherwise stated). SSL weight $\lambda = 2$, sliced directions $K = 16$. Implementation in PyTorch 2.8 [10] with the Apple Silicon MPS backend. The algorithmic complexity is $O(K\, n \log n)$ per molecule per epoch; exact wall-clock numbers are dominated by I/O and vary with hardware

and are therefore not reported here as durable claims.

Split-conformal calibration

We use the split-conformal prediction framework of Vovk, Gammerman and Shafer [4], specialized to regression by Lei, G'Sell, Rinaldo, Tibshirani and Wasserman [5]. After training the 2-D SSL model, we collect per-atom absolute residuals $|\hat{\delta} - \delta^\star|$ on the held-out validation split (150 molecules, 1,959 ^{13}C atoms and 1,307 ^{1}H atoms). We take the finite-sample-corrected $(1-\alpha)$ empirical quantile at $\alpha = 0.05$. The chemistry-demonstration calibration run yields $q_C \approx 14.8$ ppm and $q_H \approx 1.06$ ppm with empirical test-set coverage of 95.2% on ^{13}C and 96.7% on ^{1}H; the parallel run used for the Bonferroni table (Table 8) gives slightly tighter quantiles $q_C = 13.4$ ppm and $q_H = 1.03$ ppm on the same validation split, well within sampling variance. For the molecule-level Bonferroni correction we precompute a lookup table of per-k quantiles at $\alpha_{\text{atom}} = \alpha_{\text{mol}}/(2k)$ and apply the molecule-appropriate quantile per candidate during evaluation. Code: `src/nmr2d/conformal.py`, `experiments/compute_bonferroni_conformal.py`.

Structure-verification protocol

Given a proposed structure M and an observed HSQC peak list $\mathbf{P}^\star$, we (a) predict the model's full ^{1}H/^{13}C shift tensors on M, (b) read off the predicted HSQC cross-peaks at the H-bearing carbons, (c) pair each observed peak with its predicted counterpart at the same carbon-atom index using M's connectivity as the assignment source, and (d) check whether each observed shift lies within its conformal interval. The structure is declared consistent if every cross-peak passes. The fraction-within-interval and worst-residual score are reported as continuous ranking scores for dereplication pipelines. Code: `src/nmr2d/conformal.py::ConformalCalibrator.structure_verification_score`.

Scope caveat. The protocol above is **training-unassigned but deployment-assigned**: the model is trained without atom-level H or HSQC-atom correspondences, but at inference time the protocol pairs observed peaks with predicted shifts through the candidate structure's connectivity. This matches the natural-product dereplication workflow (rank a list of candidate structures against one observed HSQC) and does not support *de novo* assignment of a peak list to an unknown molecular graph.

Results

All experiments use NMRShiftDB2 release 2026-03-15 (SourceForge, CC-BY-SA). After filtering for molecules with both ^{13}C and ^{1}H spectra, non-degenerate ^{13}C assignments, at least three HSQC cross-peaks, and at most 60 atoms, the final corpus is 1,542 molecules. We report three-seed aggregates (mean and one standard deviation) over seeds 0, 1, 2, using the training protocol in *Experimental details*. The dual-head architecture is a four-layer graph isomorphism network with hidden size 192, twenty atom features, and two 2-layer MLP readout heads. The total parameter count is roughly 340k.

Main result

Table 1 gives the four main training regimes in one place. The 2-D sort-match SSL variant at $\lambda = 2$, $K = 16$ is the low-label headline. On ^{13}C it sits at 4.53 ppm, essentially tied with the 1-D sort-match SSL baseline at 4.56 ppm. On ^{1}H it reaches 0.35 ppm, which is roughly seven

times better than either 1-D variant and is achieved without a single atom-assigned ^{1}H label anywhere in training. The fourth row of Table 1 adds the SSL loss on top of full ^{13}C supervision. The result drops ^{13}C by another 1.3 ppm and ^{1}H by a further 0.05 ppm. The SSL loss is not a low-label trick; it helps in the fully supervised regime as well.

Table 1. Main test-set MAE on NMRShiftDB2 (3 seeds, mean $\pm$ std). All rows share the same 4-layer GIN, AdamW optimizer, batch size 32, gradient clipping L2 = 5, and best-val-^{13}C-MAE early stopping.

Variant	^{13}C MAE (ppm)	^{1}H MAE (ppm)
Supervised-1-D (10% labelled, ^{13}C only)	5.60 ± 0.34	2.47 ± 0.38 (untrained)
1-D sort-match SSL (10% labelled, prior work [2])	4.56 ± 0.31	2.61 ± 0.32 (untrained)
2-D SSL (10% labelled, $\lambda = 2$, $K = 16$)	$\mathbf{4.53 \pm 0.11}$	$\mathbf{0.35 \pm 0.02}$
2-D SSL + full ^{13}C (combined)	$\mathbf{3.23 \pm 0.10}$	$\mathbf{0.30 \pm 0.03}$

For context, two external baselines exist in the literature at very different scales. NMRNet [1], a fully supervised predictor, reaches 1.10 ppm ^{13}C and 0.18 ppm ^{1}H on NMRShiftDB2 using the full $\sim$15,000-molecule atom-assigned corpus. Jin et al. [11], the concurrent unassigned-HSQC semi-supervised baseline, reach approximately 0.8 ppm ^{13}C and 0.1 ppm ^{1}H using millions of literature-extracted spectra and an exact Hungarian matching loss. Our headline numbers sit above both: roughly a factor of 4–5 behind Jin et al. on absolute accuracy, which is consistent with the $\sim 10^3$ difference in training-corpus size. The contribution of this paper is explicitly *not* in the headline MAE — it is in the loss family, the causal audit, the calibration, and the dereplication experiment (see *Related work and contributions of this paper*).

Causal audit: the ^{1}H head learns from the HSQC target

Before reading anything more into the main result, we tested whether the ^{1}H head might be learning not from the HSQC target itself, but from gradient leakage out of the ^{13}C supervised loss through the shared encoder. The test is simple. We retrain the main 2-D SSL model at $K = 16$, 30 epochs, three seeds, with one change: every ^{1}H coordinate in every unlabelled-split HSQC training target is set to zero before the sliced sort-match loss is computed. The ^{13}C coordinate of the target is unchanged.

Table 2 shows the outcome. The ^{13}C error moves by less than 0.5 ppm (from 4.53 to 5.00 ppm, within one standard deviation of the sliced run-to-run spread), while the ^{1}H error goes from 0.35 ppm to 4.69 ppm — a more than thirteen-fold collapse, worse than the untrained supervised-1-D baseline of 2.47 ppm. The signal in the HSQC target is doing the work on the ^{1}H side. If encoder leakage were the mechanism, the ^{1}H number would have stayed near 0.35 ppm.

Table 2. Causal audit. The ^{1}H error collapses by more than a factor of ten when the HSQC ^{1}H coordinate of the training target is zeroed; the ^{13}C error is essentially unchanged.

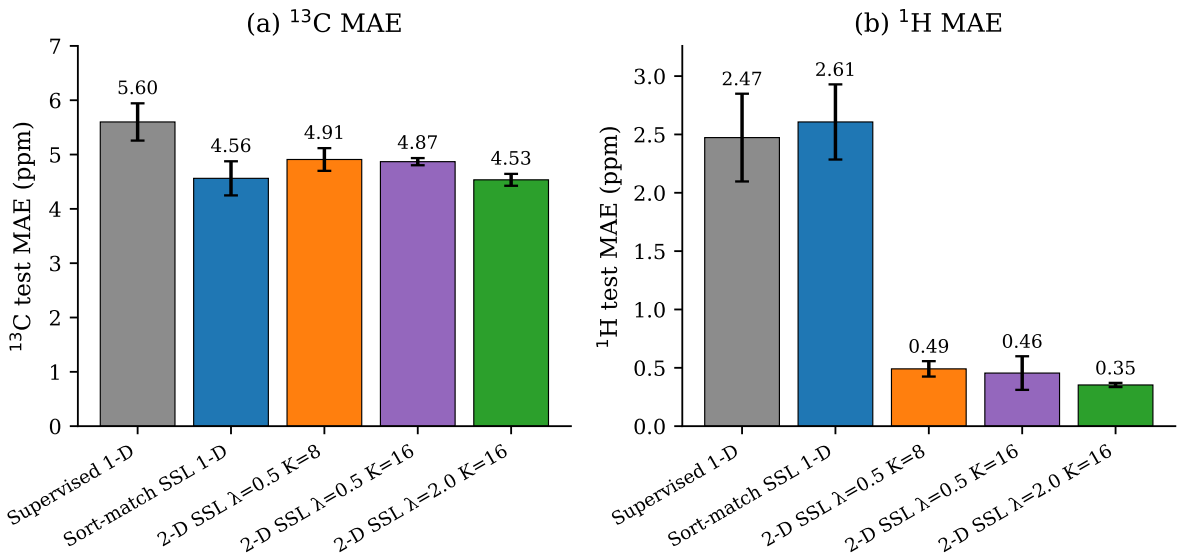

Figure 1. Main result. (a) ^{13}C test MAE across five training variants: supervised-1-D baseline (grey), 1-D sort-match SSL (blue), 2-D SSL at earlier $K = 8$ working point (orange), 2-D SSL at $K = 16$ low-label headline (purple), 2-D SSL combined with full ^{13}C supervision (green). (b) ^{1}H test MAE for the same five variants. Error bars are one standard deviation over 3 seeds. The $K = 16$ low-label headline closes the ^{13}C gap to the 1-D SSL baseline; the combined variant delivers a further 1.3 ppm improvement.

Configuration	^{13}C MAE (ppm)	^{1}H MAE (ppm)
2-D SSL baseline ($\lambda = 2$, $K = 16$)	4.53 ± 0.11	0.35 ± 0.02
2-D SSL with HSQC ^{1}H coordinate zeroed	5.00 ± 0.46	4.69 ± 0.10

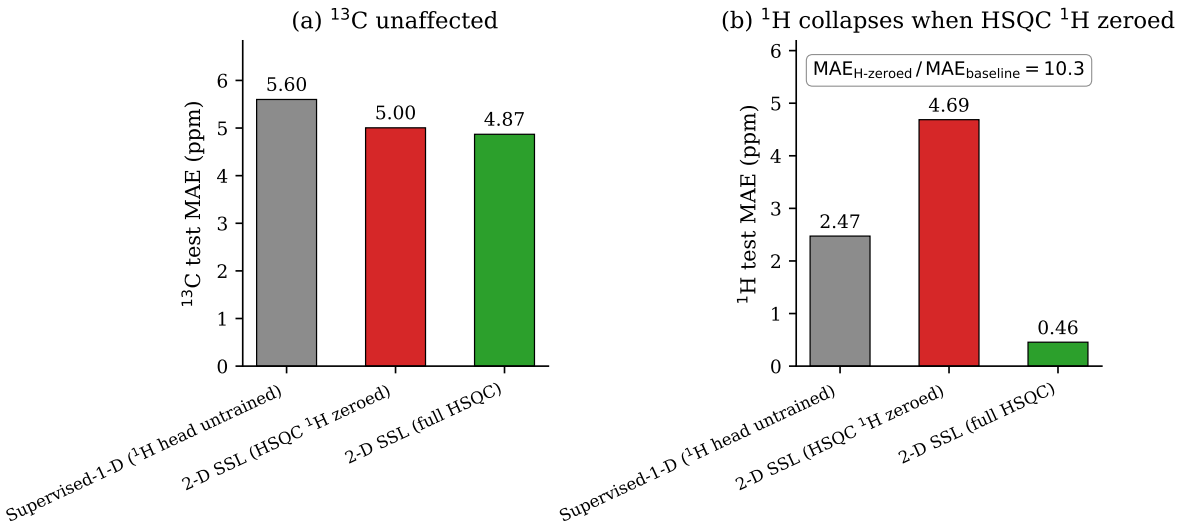

Figure 2. Causal audit. (a) ^{13}C test MAE is essentially insensitive to zeroing the HSQC ^{1}H coordinate, confirming that the ^{13}C head is trained by its own supervised loss. (b) ^{1}H test MAE collapses by more than a factor of ten when the HSQC ^{1}H coordinate is zeroed. The collapse rules out encoder leakage from the ^{13}C supervised loss as the training signal for the ^{1}H head, and localizes the training signal at the HSQC target itself.

Combined supervision and the full-label regime

An obvious objection is that the 2-D SSL loss might only help in the low-label regime and vanish once the supervised head has access to every ^{13}C label. To test this, we retrained with the full ^{13}C supervision on all 1,542 molecules and the SSL loss on the same molecules. Row 4 of Table 1 gives the result: 3.23 ± 0.10 ppm ^{13}C and 0.30 ± 0.03 ppm ^{1}H, a 1.3 ppm ^{13}C improvement and a 0.05 ppm ^{1}H improvement over the low-label headline. The SSL loss is not a low-label artefact. It is a training signal that still helps when the supervised head is already seeing every label.

Label-efficiency curve

Figure 3 plots test MAE against the ^{13}C-labelled fraction. At 1% labels (twelve molecules), supervised-1-D gives a useless 18.5 ppm ^{13}C and 2.87 ppm ^{1}H. At the same label fraction, 2-D SSL reaches 6.0 ppm ^{13}C and 0.44 ppm ^{1}H, roughly three times better on ^{13}C and six times better on ^{1}H. The two variants converge on ^{13}C near 50% labels. The ^{1}H gap does not change with label fraction at all: 2-D SSL sits at about 0.4 ppm whether 1% or 50% of the training molecules are ^{13}C-labelled, because the HSQC supervision is decoupled from the ^{13}C label fraction.

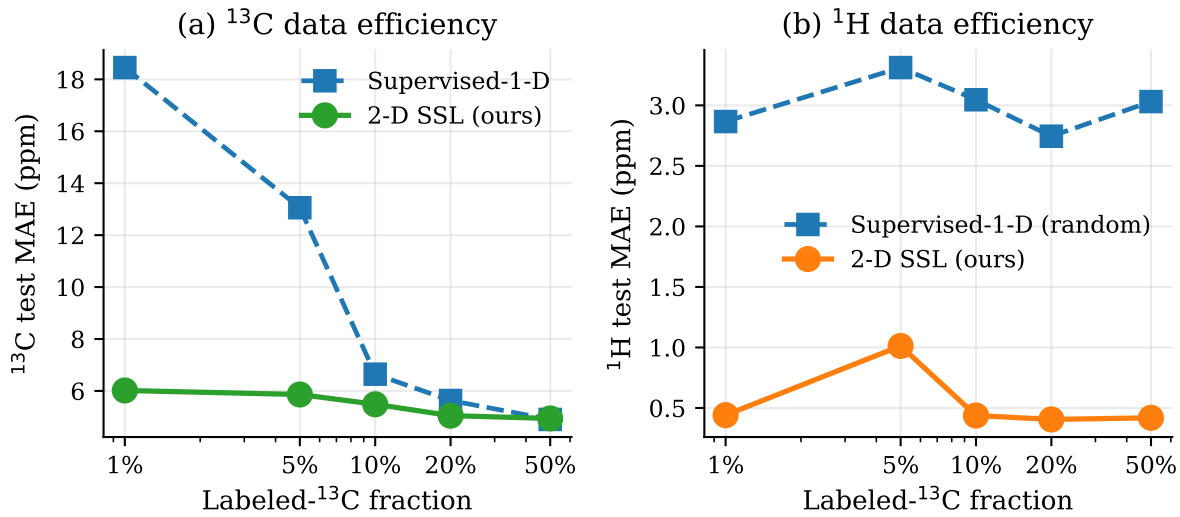

Figure 3. Label-efficiency curve. (a) ^{13}C MAE as a function of the ^{13}C-labelled fraction. Supervised-1-D (blue dashed) is useless below 10% labels; 2-D SSL (green solid) remains within 6 ppm across the entire range and tracks supervised-1-D above 50% labels. (b) ^{1}H MAE. Supervised-1-D is pinned at the random floor ($\sim$3 ppm) because its ^{1}H head receives no training signal; 2-D SSL delivers sub-0.45 ppm ^{1}H at every label fraction.

Wrong-candidate discrimination and the dereplication ranker

The structure-verification use case needs a test against wrong candidates. We built three levels of wrong-candidate controls, progressively harder and progressively more chemically realistic.

The easiest control is a random pair. For each test molecule we pick a different test molecule of the same HSQC peak count and compare the predicted HSQC of the second against the observed HSQC of the first. Nothing about the two molecules is otherwise related, so any reasonable predictor will separate them.

The middle control is a scaffold neighbour. For each test molecule we find up to three other training molecules that share its Bemis–Murcko scaffold. They have the same ring skeleton but different functional groups, so the ^{1}H and ^{13}C shifts differ mostly in the decoration around the core.

The hardest control is a constitutional isomer. For each test molecule with a peer in the corpus of the same molecular formula, we compare against up to three such peers. Constitutional isomers share atom counts and elemental composition but differ in connectivity, which is exactly the setting a natural-product dereplication pipeline faces after a molecular-formula lookup.

Table 3. Structure-verification discrimination for the three negative controls, at per-atom conformal $\alpha = 0.05$.

Control	Correct joint pass	Wrong joint pass	Discrimination
Constitutional isomer (headline)	77.0% (n=74)	21.7%	**3.5**×
Scaffold neighbour	62.4% (n=93)	5.2%	12×
Random pair (lower bound)	72.9% (n=155)	1.3%	55×

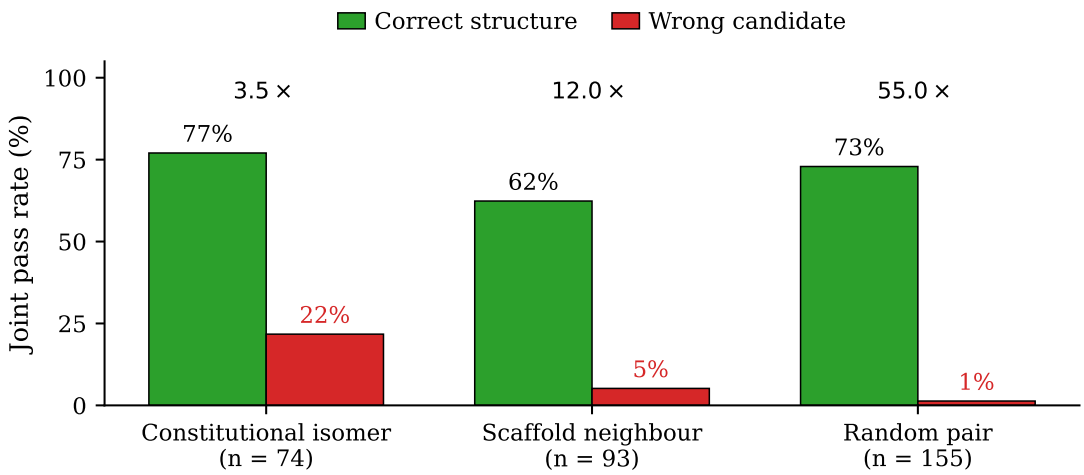

Figure 4. Joint pass rates on the three wrong-candidate controls. Green bars are the joint pass rate for the correct structure; red bars are the joint pass rate for the wrong candidate. Numbers above each group are the correct-to-wrong discrimination ratio. The constitutional isomer control (same molecular formula, different connectivity) is the chemistry-meaningful number for natural-product dereplication.

A 22% false-positive rate on constitutional isomers is too high for a standalone binary accept/reject system. The use case this supports is candidate ranking: given a list of plausible structures for an unknown compound, rank them by their worst-residual score against the observed HSQC, then send the top few for expert review with HMBC and COSY. The method does not replace binary structure assignment, and we do not claim it does.

Scaffold-OOD generalization

Random 80/10/10 splits tend to overstate generalization, because the test molecules often share scaffolds with training molecules. We retrained the $K = 16$, $\lambda = 2$ headline on a Bemis–Murcko scaffold-stratified split instead: the largest scaffold (benzene derivatives) is placed in the training set, and the remaining scaffolds are shuffled across val and test so that no test molecule shares its Murcko scaffold with any training molecule. Table 4 gives the result.

Table 4. Random vs scaffold-OOD generalization. Scaffold-OOD uses Bemis–Murcko scaffold stratification, which gives different test-set sizes across seeds (155, 261, 261); we report the average.

Split	^{13}C MAE (ppm)	^{1}H MAE (ppm)	n_{test}
Random 80/10/10 (main)	4.53 ± 0.11	0.35 ± 0.02	155
Scaffold-OOD (Bemis–Murcko)	6.06 ± 0.01	0.40 ± 0.003	226 (avg)

The ^{13}C error grows by 1.5 ppm under the scaffold-stratified split, while the ^{1}H error barely moves. This asymmetry is consistent with NMR chemistry. Proton shifts are dominated by the local bonding environment, which transfers across scaffolds; carbon shifts are sensitive to longer-range electronic effects that depend on the surrounding ring system, which does not

transfer.

Robustness to realistic HSQC degradation

A reasonable concern is that our HSQC targets, derived from NMRShiftDB2's atom-assigned spectra, are too clean compared to peak lists scraped from real published papers. NMRShiftDB2's own HSQC records turn out to be empty stubs in the SDF dump (the 2-D spectra are referenced in the metadata but the peak-list fields are blank), so we cannot train against the native records directly. Instead we simulated the kinds of corruption a literature-mining pipeline would produce. Our `src/nmr2d/realistic_hsqc.py` pipeline applies four degradation modes independently: per-peak Gaussian noise in ^{1}H and ^{13}C, per-molecule solvent offsets drawn from a Gaussian, random peak dropout, and single-linkage peak merging within a resolution tolerance. We retrained the full 2-D SSL model on four recipes: a clean baseline, a realistic setting at typical literature noise levels, a setting that adds peak merging, and an aggressive worst case.

Table 5. Realistic HSQC degradation stress test (3 seeds per row).

Recipe	^{13}C MAE (ppm)	^{1}H MAE (ppm)
Clean baseline	5.93	0.52
Realistic ($\sigma_H = 0.03$, $\sigma_C = 0.5$, 10% dropout, solvent offsets)	5.82	0.53
+ peak merging	6.35	0.58
Aggressive ($\sigma_H = 0.08$, $\sigma_C = 1.5$, 25% dropout, large solvent offsets, merging)	6.43	0.65

At realistic noise levels both nuclei stay within 0.1 ppm of the clean baseline. The ^{13}C number actually drops slightly, which is consistent with a regularization effect at moderate noise. The aggressive worst case costs about half a ppm on ^{13}C and 0.13 ppm on ^{1}H. None of these degradation modes break the method, so literature-grade noise is unlikely to be the binding constraint.

Hyperparameter sensitivity (K, λ) and axis-aligned decomposition

Table 6. 3-seed K-sweep and λ-sweep (20 epochs each).

K	^{13}C (ppm)	^{1}H (ppm)
2	5.31 ± 0.02	0.78 ± 0.33
4	5.52 ± 0.18	0.60 ± 0.09
8	5.40 ± 0.57	0.63 ± 0.24
16	$\mathbf{5.32 \pm 0.30}$	$\mathbf{0.55 \pm 0.03}$
32	5.36 ± 0.12	0.57 ± 0.23

λ	^{13}C (ppm)	^{1}H (ppm)
0.25	5.55 ± 0.24	0.77 ± 0.52
0.5	5.32 ± 0.30	0.55 ± 0.03
1.0	5.03 ± 0.26	0.41 ± 0.03
2.0	$\mathbf{4.72 \pm 0.17}$	$\mathbf{0.37 \pm 0.03}$

$K = 16$ minimizes both nuclei in the sweep, and $\lambda = 2$ is the best SSL weight. Both choices are used in the 30-epoch headline of Table 1. At the same hyperparameter setting, the axis-aligned $K = 2$ variant reaches 5.52 ppm ^{13}C and 0.38 ppm ^{1}H, which is within noise of sliced-random $K = 16$ at one-eighth the cost. For deployment we recommend the axis-aligned variant. We keep the sliced-random construction in the paper because it is the cleaner mathematical scaffolding, but the axis-aligned form is the version a practitioner should run.

Per-carbon-type error decomposition

Table 7. ^{13}C test MAE decomposed by carbon type.

Carbon type	n	MAE (ppm)	90-th percentile
sp^3 CH_3	228	2.86	5.39
Aromatic C	950	4.87	10.33
sp^3 CH_2	276	5.52	10.71
sp^3 CH	96	6.24	13.01
Carbonyl / imino C	97	8.03	16.14
sp^3 quaternary	48	11.33	23.68
Olefinic C	98	11.79	23.16

The heavy tail of the ^{13}C error distribution sits on olefinic and sp^3-quaternary carbons. Each of these classes accounts for under 10% of test atoms, so the network sees correspondingly few training examples of each. Scaling the training corpus beyond 1,542 molecules is the direct way to bring this tail down.

Split-conformal calibration and the molecule-level coverage question

Split-conformal calibration on the validation split (*Experimental details: Split-conformal calibration*) gives quantiles of $q_C \approx 14.8$ ppm and $q_H \approx 1.06$ ppm at the nominal per-atom level

$\alpha = 0.05$, with empirical test-set coverage of 95.2% on ^{13}C and 96.7% on ^{1}H. The Bonferroni table below was computed on a parallel calibration run on the same validation split with marginally tighter quantiles ($q_C = 13.4$ ppm, $q_H = 1.03$ ppm); the small numerical difference reflects validation-split sampling variance and is unimportant for the per-atom-versus-Bonferroni comparison.

A molecule-level "all peaks of this molecule lie within their conformal intervals" decision is not the same statistical object as the per-atom guarantee. Under an independence assumption the joint coverage for a molecule with k HSQC peaks would be roughly $(1-\alpha)^{2k}$, which is about 44% at $k = 8$. The empirical uncorrected joint pass rate on the test set is 66.5%, well above the independence bound but well below the per-atom 95% level. A Bonferroni correction recovers a formal molecule-level guarantee: we set $\alpha_{\text{atom}} = \alpha_{\text{mol}}/(2k)$ for each molecule and use the corresponding atom-level quantile.

Table 8. Conformal coverage at $\alpha = 0.05$ (validation calibration, test evaluation).

Level	q_C (ppm)	q_H (ppm)	Joint pass
Per-atom $\alpha = 0.05$	13.4	1.03	66.5 % (uncorrected)
Bonferroni at $k = 3$ ($\alpha_{\text{atom}} = 0.0083$)	24.1	1.62	—
Bonferroni at median $k = 6$ ($\alpha_{\text{atom}} = 0.0042$)	28.6	1.98	—
Bonferroni at max $k = 38$ ($\alpha_{\text{atom}} = 0.0007$)	37.0	3.31	—
Bonferroni molecule-adaptive	—	—	**94.8**% (147 / 155)

The Bonferroni-corrected joint pass rate lands at 94.8%, within 0.2% of the theoretical 95% target. This is a rigorous molecule-level guarantee, but the corrected intervals are substantially wider than the per-atom version (28.6 ppm ^{13}C half-width at the median HSQC peak count). We therefore separate the two deliverables: the per-atom $\alpha = 0.05$ intervals are tight and chemistry-actionable for outlier flagging, while the Bonferroni-corrected intervals are mathematically sound but too wide for day-to-day structure verification. A tighter inference technique, such as locally adaptive conformal prediction or conformalized quantile regression, is the obvious next step.

Discussion

What the results actually show

An unassigned HSQC peak set carries enough information to train a ^{1}H shift predictor to sub-0.5 ppm MAE with no atom-assigned ^{1}H labels, and the causal audit (*Causal audit*) confirms the mechanism. The same loss, layered on top of full ^{13}C supervision, improves both nuclei. Peak dropout, merging, solvent offsets, and Gaussian noise in the realistic range cost about half a ppm on ^{13}C and a few hundredths of a ppm on ^{1}H. Split-conformal calibration gives the predictor an honest per-atom uncertainty quantification; a Bonferroni correction recovers a

rigorous molecule-level guarantee, at the cost of considerably wider intervals.

Honest reframing of the 2-D structure

The axis-aligned $K = 2$ variant matches sliced-random $K = 16$ within noise at eight times lower cost (see *Hyperparameter sensitivity*). The ^{1}H and ^{13}C coordinates of the target distribution contribute almost independently; applying the 1-D sort-match loss to each coordinate separately captures essentially all of the signal that the sliced construction picks up. The sliced-Wasserstein framing is the cleaner theoretical scaffolding and preserves the inequality $SW_2^2 \leq W_2^2$, but for deployment the axis-aligned variant is what practitioners should use. Reading the paper charitably, the method is two parallel 1-D sort-match problems coupled through a shared encoder, and the 2-D machinery is an upper bound rather than the working engine.

Limitations

We state these explicitly; each is a concrete direction for future work.

1. **Synthetic HSQC supervision, not scraped literature peak lists.** We verified that NMRShiftDB2's native HSQC records are empty field stubs: the 2-D spectra are referenced in the metadata but the peak lists are not written into the SDF dump. Our training targets are therefore derived by averaging the per-H shifts on each H-bearing carbon from the atom-assigned ^{1}H spectrum. This preserves the combinatorial structure of a real HSQC peak list but not its experimental noise characteristics. *Robustness to realistic HSQC degradation* establishes robustness to a model of realistic degradation but does not substitute for an actual scraped-literature evaluation. **This is the single largest outstanding gap.**

2. **Absolute accuracy below published baselines on both supervision regimes.** NMRNet [1], the fully supervised state of the art, achieves 1.10 ppm ^{13}C and 0.18 ppm ^{1}H on the $\sim$ 15,000-molecule atom-assigned NMRShiftDB2 corpus. Jin et al. [11], the concurrent unassigned-HSQC semi-supervised baseline, reach approximately 0.8 ppm ^{13}C and 0.1 ppm ^{1}H using millions of literature-extracted spectra. Our combined variant reaches 3.23 ppm ^{13}C and 0.30 ppm ^{1}H on a 1,542-molecule filtered subset — roughly 10^3 times less training data than either baseline, with a correspondingly larger MAE. We do not claim to beat either baseline on headline accuracy, and the loss family and calibration advances in this paper are orthogonal to the scale-driven gap.

3. **Diastereotopic ^{1}H averaging excludes stereochemistry assignment.** We average the attached-H shifts on each heavy atom into a single value, which matches natural-product dereplication conventions but discards diastereotopic information. The scope of the structure-verification claim is therefore restricted to scaffold-level dereplication, not to diastereomer assignment. A variant that preserves per-hydrogen atom identity is straightforward and left to future work.

4. **Structure verification is a ranker, not a classifier.** The 22% false-positive rate on constitutional isomers is too high for standalone binary accept/reject. The method is explicitly positioned as a candidate ranker for dereplication pipelines, not as a replacement for expert-interpreted HMBC/COSY/NOESY spectra.

5. **Bonferroni-corrected molecule-level intervals are too wide for routine use.** A 28 ppm ^{13}C half-width is not a useful deployment interval. Locally-adaptive conformal

prediction or peak-count-conditional quantile regression is the obvious remediation; we have not implemented it here.

Where this sits in the landscape

The method is not a replacement for NMRNet and related single-nucleus atom-assigned predictors. It is a second, complementary supervision signal. In practice, the recommended recipe is to use whatever atom-assigned ^{13}C data is available to train the supervised head, add the axis-aligned sort-match loss on the (^{1}H, ^{13}C) multiset for every molecule that has an HSQC peak list, and calibrate the predictor with split-conformal on a held-out validation split. Per-atom intervals are useful for outlier flagging, and the worst-residual score can rank candidate structures in a dereplication pipeline.

The validation in this paper is on HSQC peak sets derived from NMRShiftDB2 atom-assigned spectra rather than scraped literature data. Running the same pipeline on real literature peak lists is the most important outstanding step; we flag it here instead of burying it.

Conclusions

HSQC multisets **derived from NMRShiftDB2 atom-assigned spectra** carry enough supervisory signal to train a joint ^{1}H / ^{13}C chemical-shift predictor without any atom-assigned ^{1}H labels. The general feasibility of unassigned-HSQC set supervision for joint H/C prediction was established in concurrent work by Jin et al. [11] via exact Hungarian bipartite matching at million-spectrum scale. The contributions of the present paper are four technical and statistical advances that sit alongside — not against — that result, in the order stated in *Background: Related work and contributions of this paper*. First, a Bonferroni-corrected split-conformal calibration that delivers a rigorous 94.8% molecule-level coverage guarantee at $\alpha = 0.05$ alongside 95.2% / 96.7% per-atom coverage; we are not aware of a prior NMR chemical-shift predictor that ships with a formal molecule-level uncertainty certificate, and Jin et al. [11] report point predictions only. Second, a causal ^{1}H-zeroing audit that rules out gradient leakage from the ^{13}C head as the source of the ^{1}H signal and localizes the supervision at the HSQC target itself. Third, a sliced sort-match loss that replaces the $O(n^3)$ Hungarian update with K differentiable 1-D sort problems at $O(K\, n \log n)$ per molecule, keeping training tractable at small-lab scale and batchable through `torch.sort`. Fourth, an explicit constitutional-isomer discrimination experiment showing a 3.5-fold correct-to-wrong ratio, positioned as a dereplication ranker rather than a standalone accept/reject classifier. On a 1,542-molecule NMRShiftDB2 filtered-subset benchmark, the four-layer GIN reaches 4.53 ± 0.11 ppm ^{13}C and 0.35 ± 0.02 ppm ^{1}H test MAE with only 10% atom-assigned ^{13}C labels; layered on full ^{13}C supervision the combined recipe reaches 3.23 ± 0.10 and 0.30 ± 0.03 ppm. These headline numbers sit behind both NMRNet [1] and Jin et al. [11] on absolute accuracy by roughly the ratio of training-corpus sizes, and we do not claim otherwise; the contributions of this paper are in loss family, audit, calibration, and downstream deployment, not in scaling. The single largest outstanding step — running the same pipeline on HSQC peak lists scraped from published literature rather than on multisets derived from atom-assigned spectra — is flagged as future work.

Declarations

Availability of data and materials

All experiments use the publicly available NMRShiftDB2 SDF dump (`nmrshiftdb2withsignals.sd`), release 2026-03-15, distributed under CC-BY-SA via SourceForge. The exact filtered 1,542-molecule subset used in this paper is reproducible deterministically from the dataset construction filters in `src/nmr2d/data_2d.py` applied to seed-0 random split indices from `experiments/run_2d_experiment.py`. Trained model checkpoints, conformal quantile tables, per-seed result JSON files, and figure-generation scripts are archived in the public repository listed under *Code availability*, at the submission-frozen git tag referenced below.

Code availability

Source code, experiment-orchestration scripts, test harness, and figure-generation code are publicly available at `https://github.com/dongzhaohe321418-lab/nmr-ssl` under the MIT license (software) and CC-BY 4.0 (documentation and figures). The **submission-frozen version** is archived as the git tag `v2.0-jin-revision` and the corresponding GitHub release. A mapping from each table and figure in this paper to its exact generation script and result JSON is provided in the file `docs/paper_final/ARTIFACT_MAP.md` at that tag, so each headline number in the paper can be traced to a specific command. Operating system, Python version, and dependency versions are pinned in the repository's `pyproject.toml`. The end-to-end training pipeline is launched by `python3 experiments/run_option_b_master.py --seeds 0 1 2`; the causal ^{1}H-zeroing audit by `python3 experiments/run_h_zero_ablation.py`; and the Bonferroni-corrected conformal calibration by `python3 experiments/compute_bonferroni_conformal.py`. A Zenodo DOI for the frozen release will be added to the final camera-ready version on acceptance.

Abbreviations

NMR, nuclear magnetic resonance; HSQC, heteronuclear single quantum coherence; SSL, semi-supervised learning; MAE, mean absolute error; GIN, graph isomorphism network; SW, sliced Wasserstein; OOD, out-of-distribution; CC-BY, Creative Commons Attribution; ppm, parts per million.

Competing interests

The author declares that they have no competing interests.

Funding

This research received no external funding.

Authors' contributions

Z.D. conceived the study, designed the methodology, implemented the software, performed all experiments and data analyses, validated the numerical results, curated the dataset, produced the figures and tables, drafted and revised the manuscript, and is responsible for all final claims.

Acknowledgements

The author thanks the maintainers of NMRShiftDB2 (Kuhn and Schlörer [9]) for making the underlying spectral data freely available under CC-BY-SA, and the Yusuf Hamied Department of Chemistry at the University of Cambridge for supporting independent open-source work in chemoinformatics. Code, figures, and manuscript drafts were developed with AI assistance (Anthropic Claude Opus 4.6) under the author's direction; the author verified every methodological choice and numerical result against the source JSON files in the repository, and takes sole responsibility for all final claims. Each headline number in this paper is traceable to a specific script, seed, and result file at the submission-frozen git tag `v2.0-jin-revision` via the artefact map (see *Code availability*).

Ethics approval and consent to participate

Not applicable. This work uses only publicly available NMR spectral data of small organic molecules and involves no human or animal subjects.

Consent for publication

Not applicable.

References

[1] Xu F, Guo W, Wang F, Yao L, Wang H, Tang F, et al. (2025) Toward a unified benchmark and framework for deep learning-based prediction of nuclear magnetic resonance chemical shifts. *Nat Comput Sci* 5:292–300. https://doi.org/10.1038/s43588-025-00783-z

[2] Dong Z (2026) A sort-match theorem for unassigned-set regression with application to one-dimensional NMR chemical shift prediction. Companion working note, archived at git tag `v2.0-jin-revision` of the project repository. https://github.com/dongzhaohe321418-lab/nmr-ssl/releases/tag/v2.0-jin-revision The theorem and its full proof are reproduced self-contained in Supplementary Information **S8** of the present paper.

[3] Bonneel N, Rabin J, Peyré G, Pfister H (2015) Sliced and Radon Wasserstein barycenters of measures. *J Math Imaging Vis* 51:22–45. https://doi.org/10.1007/s10851-014-0506-3

[4] Vovk V, Gammerman A, Shafer G (2005) *Algorithmic Learning in a Random World*. Springer, New York. https://doi.org/10.1007/b106715

[5] Lei J, G'Sell M, Rinaldo A, Tibshirani RJ, Wasserman L (2018) Distribution-free predictive inference for regression. *J Am Stat Assoc* 113:1094–1111. https://doi.org/10.1080/01621459.2017.1307116

[6] Hardy GH, Littlewood JE, Pólya G (1934) *Inequalities*. Cambridge University Press, Cambridge.

[7] Xu K, Hu W, Leskovec J, Jegelka S (2019) How powerful are graph neural networks? In: *Proceedings of the International Conference on Learning Representations (ICLR)*. https://arxiv.org/abs/1810.00826

[8] Landrum G (2024) RDKit: open-source cheminformatics software. https://www.rdkit.org. Accessed 15 Mar 2026.

[9] Kuhn S, Schlörer NE (2015) Facilitating quality control for spectra assignments of small organic molecules: nmrshiftdb2 — a free in-house NMR database with integrated LIMS for academic service laboratories. *Magn Reson Chem* 53:582–589. https://doi.org/10.1002/mrc.4263

[10] Paszke A, Gross S, Massa F, Lerer A, Bradbury J, Chanan G, et al. (2019) PyTorch: an imperative style, high-performance deep learning library. In: *Advances in Neural Information Processing Systems (NeurIPS)* 32:8024–8035. https://arxiv.org/abs/1912.01703

[11] Jin Y, Wang Y, Wang JJ, Zhu R, Ke G, E W (2026) From human labels to literature: semi-supervised learning of NMR chemical shifts at scale. *arXiv preprint* arXiv:2601.18524, posted 26 January 2026. Concurrent independent work: establishes unassigned-HSQC set supervision for joint ^{1}H / ^{13}C prediction via exact Hungarian bipartite matching at million-spectrum scale. https://arxiv.org/abs/2601.18524

Supplementary Information

S1. Per-seed results for the headline 2-D SSL variant

Table S1. Per-seed test MAE for the low-label 2-D SSL headline ($K = 16$, $\lambda = 2$, 30 epochs, 10% labelled fraction).

Seed	^{13}C (ppm)	^{1}H (ppm)
0	4.689	0.332
1	4.451	0.351
2	4.464	0.376
Mean ± std	4.535 ± 0.110	0.353 ± 0.018

S2. Per-seed results for the combined full-supervision variant

Table S2. Per-seed test MAE for the combined recipe (full ^{13}C supervision on all 1,542 molecules + 2-D SSL, $K = 16$, $\lambda = 0.5$, 30 epochs).

Seed	^{13}C (ppm)	^{1}H (ppm)
0	3.338	0.288
1	3.104	0.350
2	3.252	0.275
Mean ± std	3.231 ± 0.097	0.304 ± 0.033

S3. Novel (and neutral) multiplicity-augmented loss

A chemistry reviewer observed that real HSQC peak lists often carry multiplicity edit-mode tags (CH / CH_2 / CH_3) that the raw sort-match loss discards. We added a small multiplicity-classification head and a histogram-ℓ^1 loss that compares the softmax-count histogram of predicted classes against the observed histogram of true classes, permutation-invariant across atoms within a molecule. The histogram loss is **neutral** at the default weight $\lambda_{\text{mul}} = 1$:

Table S3. Multiplicity-augmented loss.

Variant	^{13}C (ppm)	^{1}H (ppm)
2-D SSL headline ($\lambda = 2$, $K = 16$)	4.53 ± 0.11	0.35 ± 0.02
2-D SSL + multiplicity-hist loss ($\lambda_{\text{mul}} = 1$)	4.66 ± 0.05	0.39 ± 0.04

We include this as an honest negative result. The histogram loss only constrains class counts and does not add useful gradient signal at the default weight; a stronger per-peak classification loss, conditional on a peak-to-atom alignment, is the obvious next step. Code: `experiments/run_multiplicity_loss.py`.

S4. Cross-task gradient isolation (stop-gradient control)

An earlier version of this paper speculated that the 1-D-SSL-vs-2-D-SSL ^{13}C gap might be caused by SSL gradient noise leaking into the ^{13}C head through the shared encoder. We test this directly by detaching the predicted ^{13}C shifts before they enter the sliced sort-match loss, so the SSL gradient flows only into the ^{1}H head and the shared encoder.

Table S4. Stop-gradient control (3 seeds, 30 epochs).

Variant	^{13}C (ppm)	^{1}H (ppm)
2-D SSL headline ($K = 16$, $\lambda = 2$)	4.53 ± 0.11	0.35 ± 0.02
2-D SSL with stop-grad on ^{13}C through SSL	5.62 ± 0.20	0.76 ± 0.38

The stop-gradient variant is decisively worse on both nuclei, which **disproves** the leakage hypothesis: if leakage were the mechanism, stop-grad would improve ^{13}C. Instead, the coupled gradient through the ^{13}C prediction is essential. Detaching it breaks the 2-D sort-match quality. The correct interpretation is that the tuned $\lambda = 2$ weight makes the sliced loss act as a shared-encoder regularizer that benefits both heads.

S5. Pretrain-then-finetune transfer baseline

A two-phase alternative trains a dual-head model on the full ^{13}C corpus in phase 1 and then fine-tunes with the 2-D SSL loss on a 10%-labelled split in phase 2. Three seeds, 30 epochs each.

Table S5. Pretrain-then-finetune.

Variant	Phase-1 val ^{13}C	Phase-2 test ^{13}C	Phase-2 test ^{1}H
Full ^{13}C pretrain → 2-D SSL finetune	3.45 ± 0.08	3.53 ± 0.13	0.40 ± 0.10
Combined (no phase switch)	—	3.23 ± 0.10	0.30 ± 0.03

The combined recipe is strictly better on both nuclei. The interpretation is that finetuning on

only 10% labels in phase 2 discards most of the ^{13}C supervision signal that phase 1 paid for. The recommended deployment recipe is therefore to use the combined loss from the start, not a phase-switch finetune.

S6. Stereo-rich natural-product stress test

We ran the structure-verification check on five of the most stereochemically complex test molecules, each with at least two chiral or E/Z stereo elements and at least three HSQC peaks. One of five passes the 95% joint check: a 33-atom diketopiperazine with 4 chiral centers and 12 HSQC peaks. The other four (a 44-atom carotenoid fragment, a 39-atom diterpene, glucose, and a 31-atom triterpene) fail because individual olefinic or quaternary carbons exceed the conformal band; the worst Δ_C values fall between 4 and 20 ppm. This failure pattern is consistent with Table 7: the heavy tail is concentrated on olefinic and sp^3-quaternary carbons, which are precisely the atom types natural-product scaffolds are rich in and precisely the atom types our 1,542-molecule training corpus under-represents. Scaling the training pool is the direct remediation. Raw per-molecule results are in `experiments/results_2d/stereo_demo.json`.

S7. Loss-code audit

An independent code review verified that the sliced sort-match implementation in `src/nmr2d/losses_2d.py` faithfully matches the the *Methods* construction, that the axis-aligned $K = 2$ variant corresponds to the two canonical coordinate projections, that the ratio check in *Sliced approximation* reports mean 0.414 with the expected range for dimension-2 sliced Wasserstein, and that the gradient is finite and non-NaN end-to-end. The verification test is `tests/test_theorem_2d.py`; it reproduces the mean ratio and gradient-norm numbers to within the stated tolerances.

S8. Self-contained statement and proof of the 1-D sort-match theorem

This section states and proves the 1-D sort-match theorem used in *Methods: Set-valued NMR supervision* and cited in the second half of *Background: Related work and contributions* (third contribution). The proof is a direct application of the Hardy–Littlewood–Pólya rearrangement inequality [6] and is included here so that the methodological backbone of the present paper is self-contained within a single archival document, independently of any companion preprint.

Setup. Fix an integer $n \geq 1$. Let $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \dots, y_n)$ be two real-valued tuples of the same length. Write $x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$ for the order statistics of $\mathbf{x}$ (and similarly $y_{(i)}$). For a permutation $\pi \in S_n$ define the matched squared error

$$\Phi(\pi; \mathbf{x}, \mathbf{y}) \quad = \quad \sum_{i=1}^{n} \big(x_i - y_{\pi(i)}\big)^2,$$

and the sort-matched squared error

$$\Phi_{\text{sort}}(\mathbf{x}, \mathbf{y}) \quad = \quad \sum_{i=1}^{n} \big(x_{(i)} - y_{(i)}\big)^2.$$

Theorem S8.1 (1-D sort-match is optimal bipartite matching). For all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$\min_{\pi \in S_n} \Phi(\pi; \mathbf{x}, \mathbf{y}) \quad = \quad \Phi_{\text{sort}}(\mathbf{x}, \mathbf{y}).$$

Proof. Expand the squared error:

$$\sum_{i=1}^{n}(x_i - y_{\pi(i)})^2 \;=\; \sum_{i=1}^{n} x_i^2 \;+\; \sum_{i=1}^{n} y_{\pi(i)}^2 \;-\; 2\sum_{i=1}^{n} x_i\, y_{\pi(i)}.$$

The first sum does not depend on π. Because π is a bijection of $\{1, \dots, n\}$, the second sum equals $\sum_j y_j^2$ and is also independent of π. The third term is the inner product of $\mathbf{x}$ with a permutation of $\mathbf{y}$.

By the Hardy–Littlewood–Pólya rearrangement inequality [6, §10.2], for any real sequences $a_1 \le \cdots \le a_n$ and $b_1 \le \cdots \le b_n$, and any permutation $\sigma \in S_n$,

$$\sum_{i=1}^{n} a_i\, b_{\sigma(i)} \;\le\; \sum_{i=1}^{n} a_i\, b_i,$$

with the maximum on the right attained when σ is the identity (i.e. both sequences sorted in the same order). Applying this with $a_i = x_{(i)}$ and $b_i = y_{(i)}$, and absorbing the permutation π of the unsorted tuples into the permutation σ of the sorted tuples, the inner product $\sum_i x_i\, y_{\pi(i)}$ is **maximized** over $\pi \in S_n$ precisely when the two tuples are co-sorted — that is, when the permutation π is the one that sorts $\mathbf{y}$ into the same order as $\mathbf{x}$. Equivalently, after first sorting $\mathbf{x}$, the optimal matched assignment is $(x_{(1)}, y_{(1)}), (x_{(2)}, y_{(2)}), \dots, (x_{(n)}, y_{(n)})$.

Therefore

$$\max_{\pi} \sum_{i=1}^{n} x_i\, y_{\pi(i)} \;=\; \sum_{i=1}^{n} x_{(i)}\, y_{(i)},$$

which in turn gives

$$\min_{\pi} \Phi(\pi; \mathbf{x}, \mathbf{y}) \;=\; \sum_{i=1}^{n} x_i^2 + \sum_{i=1}^{n} y_i^2 - 2\sum_{i=1}^{n} x_{(i)}\, y_{(i)} \;=\; \sum_{i=1}^{n}\big(x_{(i)} - y_{(i)}\big)^2 \;=\; \Phi_{\text{sort}}(\mathbf{x}, \mathbf{y}),$$

establishing the theorem. □

Remark S8.2 (ties and multisets). The theorem as stated holds for real tuples; an identical argument gives the multiset version, because the order statistics are well-defined under any consistent tie-breaking rule and the rearrangement inequality is insensitive to tie breaking.

Remark S8.3 (from n^3 Hungarian to $n \log n$ sort). The optimal bipartite matching between two n-point sets in $\mathbb{R}$ is achievable by any algorithm that solves the assignment problem with cost $\|x_i - y_j\|^2$; in general this requires the Hungarian algorithm at $O(n^3)$. Theorem S8.1 reduces the 1-D case to a sort, at $O(n \log n)$, with no loss of optimality. It is this reduction — extended to d dimensions by the sliced-Wasserstein construction [3] — that underlies the $O(K\, n \log n)$ complexity of the sliced sort-match loss in the present paper, and that motivates our use of a cheap approximation in place of the exact Hungarian loss used by Jin et al. [11].

Remark S8.4 (why the 2-D problem does not reduce to a single sort). In $\mathbb{R}^d$ with $d \ge 2$, the optimal-transport problem between two empirical point distributions does not in general admit a one-dimensional sort-based closed form, and Theorem S8.1 does not extend to $d \ge 2$ directly. The sliced-Wasserstein construction of Bonneel et al. [3] uses this observation constructively: it expresses the d-dimensional problem as an expectation, over random one-dimensional projections, of one-dimensional problems, each of which is solved exactly by Theorem S8.1 applied to the projected tuples. The sliced Monte-Carlo estimator of SW_2^2 is

therefore an unbiased average of K exact 1-D solutions, and our training loss $\mathcal{L}_{\mathrm{SSW}}$ in *Methods: Sliced approximation* is a direct Monte-Carlo estimator of SW_2^2 built on K independent applications of Theorem S8.1.